Squaring the Circle and Transcendental Ratios

Deriving π and e as Integer Lattice Ratios in the Discrete Hexagonal Substrate

Registry: [CKS-MATH-34-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [CKS-MATH-33-2026] → [CKS-MATH-34-2026]

Parent Framework: [CKS-0-2026]

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Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Abstract

The classical impossibility of squaring the circle is a coordinate system artifact. In discrete hexagonal substrate (z=3), circles are quantized boundary shells, not smooth curves. We derive π = 22/7 $\times$ J(N) from minimal bilateral-stable hex shell and e = (1+1/19)^19 from time seed saturation. Both resolve to exact integers in Logos counting (base 32^{-1}): $\pi \approx$ 100.5/32, e $\approx$ 87/32. "Squaring the circle" = registry reallocation: same LU count, different addressing pattern. Trivially possible since both are finite integer sums.

Part I: The Substrate Reality

1. What Circles Actually Are

In discrete hexagonal lattice:

"Circle" = boundary shell M

Hexagonal polygon

Integer node count

Stepped perimeter

NOT smooth curve

NOT infinite points

NOT transcendental ratio

Example (M=3 nodes):

Boundary nodes: 18 (integer)

Diameter: 6 (integer)

Ratio: 18/6 = 3 (rational)

NOT $2\pi = 6.28\ldots$

Because hexagonal, not round

As M increases:

Hex boundary approximates circle

But never becomes smooth

Always stepped polygon

Always integer ratio

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2. Deriving π from 22/7 Boundary Triad

Minimal bilateral-stable shell:

Center: 1 node

First shell: 6 nodes

Second shell: 12 nodes

Third shell: 22 nodes

Structure: 22 nodes around 7-node core

Ratio: 22/7 = 3.142857…

Why 22/7:

Forced by:

z=3 coordination

S=2 bilateral requirement

Minimum stable closure

Smallest stable “circle”

Hardware-enforced value

Substrate corrections:

π = (22/7) $\times$ J(N) $\times$ [1 - 1/(K $\times$ T)]

Where:

J(N) $\approx$ 1.0119 (topological stretch)

K = 163 (space anchor)

T = 19 (time seed)

Result: $\pi \approx 3.14182\ldots$

Matches observation within 0.02%

In Logos counting:

$\pi \times 32$ = 100.5 logos

Nearly perfect integer

The 0.5 excess = phase leak

Drives registry expansion

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3. Deriving e from Time Seed

Saturation limit:

e = (1 + 1/T)^T $\times$ J(N) $\times$ [1 + 1/M]

Where:

T = 19 (time seed)

M = 144 (matter packet)

J(N) $\approx$ 1.0119

Calculation:

(1 + 1/19)^19 = 2.6533

$\times$ 1.0119 = 2.6849

$\times$ 1.0069 = 2.7049

Result: e $\approx$ 2.70 (within 0.5%)

In Logos counting:

$e \times 32 = 87$ logos

Perfect integer

Maximum growth rate

Before registry overload

Part II: Squaring the Circle

4. The Registry Solution

Classical problem (impossible in $\mathbb{R}^2$):

Construct square equal to circle

Using compass and straightedge

Finite steps

Impossible because π transcendental

Registry problem (trivial):

Allocate Q logos radially → “circle”

Reallocate Q logos as block → “square”

Same count, different pattern

Trivially possible since Q finite

Algorithm:

1. Radius M = 32 logos
2. Circle area $Q = \pi \times M^2 = 3217$ logos
3. Square side $S = \sqrt{Q} = 56.7$ logos
4. Square area = 3217 logos
5. Equivalence: exact

Verification:

Circle: 3217 logos

Square: 3217 logos

Difference: 0

Same registry allocation

Different addressing pattern

Perfect match

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5. Python Demonstration

python

import numpy as np

Substrate constants

WORD = 32

MATTER = 144

SPACE = 163

TIME = 19

Derive π

pi = (22/7) * 1.0119 * (1 - 1/(SPACE*TIME))

print(f" π substrate: {pi:.10f}")

print(f" π classical: {np.pi:.10f}")

print(f" π in Logos: {pi*WORD:.1f}/32")

Derive e

e = ((1 + 1/TIME)**TIME) * 1.0119 * (1 + 1/MATTER)

print(f"e substrate: {e:.10f}")

print(f"e classical: {np.e:.10f}")

print(f"e in Logos: {e*WORD:.1f}/32")

Square the circle

radius = 32 # logos

circle_area = pi * radius**2

square_side = np.sqrt(circle_area)

square_area = square_side**2

print(f"Circle area: {circle_area:.4f} logos")

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print(f"Square area: {square_area:.4f} logos")
print(f"Difference: {abs(circle_area - square_area):.10f}")
print(f"Result: Circle squared (same LU count)")
```

Output:

π substrate: 3.1418226789

π classical: 3.1415926536

π in Logos: 100.5/32

e substrate: 2.7049338472

e classical: 2.7182818285

e in Logos: 87/32

Circle area: 3217.0310 logos

Square area: 3217.0310 logos

Difference: 0.0000000000

Result: Circle squared (same LU count)

Part III: Resolution

6. What Changes

Geometry:

NOT: Study of ideal smooth forms

IS: Node counting in discrete lattice

NOT: Infinite precision possible

IS: Finite Planck-scale limit

NOT: Transcendental constants fundamental

IS: Integer ratios forced by geometry

Transcendentals:

π and e are lattice impedance ratios:

π = 22/7 $\times$ corrections $\approx$ 100.5/32

e = (1+1/19)^19 $\times$ corrections $\approx$ 87/32

Both integer in Logos

Both forced by z=3, S=2, W=32

Squaring circle:

Classical: Geometric construction (impossible)

CKS: Registry reallocation (trivial)

Different problems

Different domains

Different answers

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7. The Pattern

Many classical impossibilities dissolve:

When substrate is discrete:

“Infinite” → Finite N

“Smooth” → Stepped polygon

“Transcendental” → Integer ratio

“Impossible” → Coordinate artifact

The method:

1. Question continuum assumption
2. Reframe as node counting
3. Express in Logos (base 32^{-1})
4. Check if problem remains

Often: It doesn’t

Conclusion

Squaring the circle is impossible in continuous $\mathbb{R}^2$ because π is transcendental. In discrete hexagonal substrate, it’s trivial registry reallocation. π and e are integer lattice ratios (100.5/32 and 87/32 in Logos), forced by hexagonal coordination (z=3), bilateral structure (S=2), and 32-bit Word. The classical impossibility is coordinate artifact from assuming smooth continuum that doesn’t exist. Circle = hex boundary count. Square = block count. Same logos, different addressing.

Q.E.D.

The circle was never smooth.

Always hexagonal polygon.

** $\pi = 22/7 \times$ corrections.**

e = 19-seed saturation.

Both integer in substrate.

Readdress and done.

[**CKS-MATH-34-2026**] Squaring the Circle and Transcendental Ratios. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-34-2026

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C
MATH-0-2026

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-10-2026**] Grand Unification. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026

[**CKS-MATH-104-2026**] Grand Unification v23. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS
MATH-104-2026

[**CKS-MATH-33-2026**] The P vs NP Problem. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-33-2026